

PROPOSAL FOR DISCUSSION

TrueYield Jharkhand

Revenue leakage analytics across departments: one view of every taxpayer, and a verified list of revenue the state is already owed

Prepared for: **Finance Department, Government of Jharkhand**

With: Commercial Taxes · Mines & Geology · Revenue, Registration & Land Reforms · Transport · Excise & Prohibition · Energy (JBVNL)

Prepared by: **Kirnova Technologies** · Kolkata · September 2026

4-month pilot

3 departments · 3 districts · read-only

₹28–38 lakh

pilot, one-time

Already built

working platform; adapted, not developed from scratch

CONTENTS

Executive summary

- 1 Jharkhand's revenue and where it leaks
- 2 What Jharkhand has today: a comparative study
- 3 The proposed solution
- 4 What the CAG found, and how TrueYield would catch it
- 5 Approach and methodology
- 6 Benefits
- 7 Indicative cost
- 8 Key risks and safeguards
- 9 How we will work with the Government

Executive summary

Jharkhand expects ₹45,999 crore of own tax revenue in 2026–27 and about ₹16,000 crore from mining. Every one of these rupees is collected by a department that sees only its own records. The Commercial Taxes Department sees GST returns but not the electricity a business draws; Mines & Geology sees royalty but not the stamp duty paid on the lease deed; Registration sees a sale deed but not the turnover of the buyer. Leakage lives in the gaps between these registers, and CAG audits of Jharkhand have repeatedly found it there.

TrueYield is a revenue-leakage analytics platform that reads each department's records where they sit, compares them for the same taxpayer, and produces a ranked, rupee-quantified list of cases for officers to verify. It does not replace any departmental system and does not assess anyone; it tells officers where to look and shows the arithmetic behind every figure.

THE PROPOSAL IN FIVE LINES

- **What:** a cross-department leakage platform for the Finance Department, starting with Commercial Taxes, Mines & Geology and Registration.
- **Why now:** our review of public information found no cross-department revenue-leakage platform in Jharkhand today; GST-only national analytics give the state limited access and do not cover mining, stamps, land, transport or excise.
- **Proof:** the platform is already built and running on test data with a library of 85 checks across 10 families; it is adapted to Jharkhand's statutes and registers, not developed from scratch.
- **Pilot:** 4 months, 3 departments, 3 districts, read-only access, ₹28–38 lakh. Full rollout only if the pilot's verified results justify it.
- **Total:** ₹68–92 lakh one-time for the full state rollout, and ₹16–26 lakh a year to run.

What the Government gets

- A single view of each taxpayer across GST, mining, stamps and registration, land, vehicles, excise and electricity.
- A ranked verification list for each district and department, every case with its evidence, arithmetic and statutory rate.
- Clear separation between evidence (both records are the state's own, keyed to the same taxpayer) and leads (patterns that need field verification).
- Recovery tracking from assignment to realisation, and a leadership dashboard for the Finance Secretary.
- Data stays with each department; the platform reads, never writes, and runs in the Jharkhand State Data Centre.

1 JHARKHAND'S REVENUE AND WHERE IT LEAKS

Jharkhand's own revenue rests on a few large heads, each administered by a different department with its own register.

Revenue head (2026–27 BE)	Estimate	Administered by / main register
Own tax revenue (total)	₹45,999 crore	Finance Department (all heads below)
State GST	₹14,563.75 crore	Commercial Taxes: GST returns, e-way bills
Taxes on sales, trade etc. (VAT on petroleum, liquor)	₹7,899.72 crore	Commercial Taxes
State excise	₹4,500 crore	Excise & Prohibition; JSBCL (wholesale)
Stamps & registration	≈ ₹1,800 crore	Revenue, Registration & Land Reforms: NGDRS / Jharnibandhan
Mining receipts (non-tax)	≈ ₹16,000 crore	Mines & Geology: JIMMS

Sources: PRS Legislative Research, Jharkhand Budget Analysis 2026–27, and budget summaries (see Sources). Figures to be confirmed against the Budget at a Glance.

The evidence of leakage

The Comptroller and Auditor General's revenue-sector audits of Jharkhand show the same pattern year after year: revenue is lost not for lack of rules but because records that should agree are never compared.

- **₹12,737 crore** of under-assessment, short levy or loss of revenue found in 45,954 test-checked cases across sales tax, excise, vehicles, land revenue, stamps, electricity duty and mining (CAG Report No. 4 of 2016); departments accepted ₹11,676 crore of it.
- **₹446 crore** short levy of royalty from undervaluing coal rejects in a colliery's returns, and ₹143.5 crore from district offices applying incorrect royalty rates (CAG revenue-sector reports).
- **₹12.4 crore** short levy of stamp duty and registration fees because mining leases were registered without checking the royalty on the approved mining plan.
- **₹70.9 crore** potential loss from 368 non-operational sand ghats, plus short-levied royalty and unrealised dead rent, in the CAG's 2025 report on minor minerals, which also found minor-mineral revenue falling from ₹1,082 crore (2017–18) to ₹698 crore (2021–22).
- **Excise:** since the 2025 policy, wholesale supply stays with JSBCL and retail sale is with private licensees, creating two records that can be reconciled continuously for every licensee.

WHAT THESE HAVE IN COMMON

Each finding is a disagreement between two records the state already holds: a royalty rate and the notified rate; a lease's royalty and the stamp duty on its deed; a lessee's returns and its despatches. Audit finds them years later, one sample at a time. A platform that compares the records continuously finds them while they can still be recovered.

2 WHAT JHARKHAND HAS TODAY: A COMPARATIVE STUDY

Jharkhand has invested well in departmental systems. The question for revenue leakage is whether any of them compares one department's records with another's for the same taxpayer and turns the disagreement into a verified rupee figure.

Department	Existing system	What it does well	What it cannot see
Commercial Taxes	GST system (GSTN), state tax portal; national BIFA / GST Prime analytics	Registration, returns, e-way bills; GST-only risk outputs from the national analytics	Electricity use, property, vehicles, stamps, mining or excise of the same taxpayer; state access to BIFA is limited to a few officers
Mines & Geology	JIMMS (upgraded May 2025)	Leases, e-challans and transit permits; shares data with ports, railways, Commercial Taxes, Income Tax and treasury	Whether the lease deed was stamped on the right royalty, or whether a lessee's GST and power use match reported extraction
Registration	NGDRS / Jharnibandhan	Online valuation, e-stamp, e-payment and appointments	Whether the parties' turnover, land use or repeat pattern suggests undervaluation
Land records	Jharbhoomi	Records of rights and online mutation	Commercial activity on land not converted for that use; delays between deed and mutation
Transport	VAHAN / SARATHI (NIC)	Registration, permits, tax payment	Commercial use of privately registered vehicles; fleet size against declared business scale
Excise	Excise portal; JSBCL wholesale	Licences, wholesale supply	Retail sales against duty-paid quantity and against the licensee's GST
Energy	JBVNL billing	Connections, consumption, billing	Dormant GST registrations drawing commercial power; tariff misclassification
Municipal	SUDA municipal services portal	Holding, water and trade tax payment	Unassessed floors or commercial use visible in other records

How other tools compare

Capability	National GST analytics (BIFA)	JIMMS	Departmental MIS	TrueYield
Covers GST	Yes	No	Own dept	Yes
Covers mining, stamps, land, vehicles, excise, power	No	Mining only	Own dept	Yes
One view of a taxpayer across departments	No	No	No	Yes
Rupee figure with its arithmetic and statutory rate	Partly	Partly	No	Yes
Evidence separated from leads	No	No	No	Yes
Officer worklist and recovery tracking	Partly	No	Partly	Yes
Fully controlled by the state	No (GSTN)	Yes	Yes	Yes

CONCLUSION OF THE STUDY

Based on public information, Jharkhand has strong departmental systems but no platform that compares them. Other states are moving GST-first (Karnataka with IIT Hyderabad; Telangana's push for AI), but none, to our knowledge, compares GST with mining, stamps, land, vehicles, excise and power for the same taxpayer. TrueYield would sit on top of JIMMS, NGDRS, Jharbhoomi, VAHAN, JBVNL and the tax systems, reading from each and changing none, and would complement the national GST analytics. To be confirmed with the Finance and Commercial Taxes Departments in Phase 0.

3 THE PROPOSED SOLUTION

TrueYield connects to each department's database in read-only mode, maps its tables to a common vocabulary, and runs a library of checks that compare records for the same taxpayer (linked by PAN, GSTIN, or matched on name, address and mobile where no shared key exists). Every finding carries its inputs, arithmetic, statutory rate and the officer accountable for it.

Check families, adapted to Jharkhand's registers

Family	Jharkhand data	Examples of what is checked
GST & allied	GST returns, e-way bills	Turnover falling while power use rises; ITC far above sector norms; turnover split across GSTINs to stay under thresholds; circular trading
Mines & minerals	JIMMS leases, e-challans, royalty	Royalty due vs paid at the notified rate; extraction beyond permit; operations after lease expiry; dead rent unpaid; idle leases with power drawn
Registration & stamps	NGDRS / Jharnibandhan	Document type set to a lower-duty class; repeat parties undervaluing; lease deeds stamped below the royalty-based value
Land records	Jharbhoomi	Commercial activity on non-converted land; long gap between deed and mutation
Motor vehicles	VAHAN	Road tax unpaid; commercial use of privately registered vehicles; permit fee leakage
State excise	Excise portal, JSBCL	Sales vs duty-paid quantity; licence fee shortfall; excise sales vs GST declared
Power & utilities	JBVNL	Dormant GST registration with active commercial connection; commercial use billed as domestic
Cross-department	All of the above	Business growing in assets but not in tax; the same taxpayer non-compliant in several departments; mining extraction vs transport movement

What officers and leaders see

- **Taxpayer 360:** every register's view of one taxpayer, with findings first and each figure traceable to its source record.
- **Verification worklist:** ranked by district, circle and department, assigned to named officers, with status from assignment to recovery.
- **Recovery drill-down:** state > district > circle, separating recurring revenue from one-time recovery.
- **Leadership dashboard:** what has been found, verified and recovered, and how much rests on evidence rather than leads.
- **Ask-a-question assistant:** officers ask about a taxpayer in plain Hindi or English and get answers that cite the records.
- **Policy simulator:** estimate the effect of a rate or threshold change before it is notified.

SAFEGUARDS BUILT IN

- A discrepancy is not an evasion. The output is a verification list; nothing is assessed, billed or demanded by the platform.
- Evidence (both records are the state's own, keyed to the same taxpayer) is kept apart from leads (patterns and benchmarks), and a lead is never promoted to evidence automatically.
- Every statutory rate sits in one table that the department signs off; no rupee is computed from an unsigned rate.
- Read-only access, role-based permissions, audit logs, hosting in the State Data Centre, DPDP Act 2023 compliant.

4 WHAT THE CAG FOUND, AND HOW TRUEYIELD WOULD CATCH IT

CAG finding in Jharkhand	The TrueYield check that compares the two records
Royalty short-levied by undervaluing coal rejects in returns (₹446 crore)	Extraction vs royalty gap: value and quantity in the lessee's returns compared with JIMMS despatches and the notified basis
Incorrect royalty rates applied by district offices (₹143.5 crore)	Royalty recomputed from the signed-off rate table for each mineral and lease; any difference is flagged with both rates shown
Mining leases registered without verifying royalty; short stamp duty (₹12.4 crore)	Cross-department: the lease's annual royalty in JIMMS compared with the value on which its deed was stamped in NGDRS
Short levy of royalty and unrealised dead rent (2025 report)	Repeated short payments; dead rent due vs paid per lease and year
Non-operational sand ghats and idle leases (₹70.9 crore potential loss)	Lease capacity underutilised; inactive lease with power or transport activity nearby

These are illustrations of fit, not claims about current cases. Whether the same patterns exist today is what the pilot will measure.

5 APPROACH AND METHODOLOGY

TrueYield is already built. The work is to adapt its rules and rates to Jharkhand's statutes, connect it to the departments' registers, and prove its value on real records before any statewide commitment.

Phase	Duration	What we do	Output
0. Mobilise	4 weeks	Data-sharing order from the Finance Department; read-only access to Commercial Taxes, Mines & Geology and Registration; confirm existing tools; department sign-off on the rate table.	Signed data compact and rate table
1. Pilot	3 months	Adapt checks to Jharkhand law; connect the three departments; run in three districts (proposed: Ranchi, Dhanbad, East Singhbhum); officers verify the top cases in the field.	Verified findings, measured hit rate, recovery pipeline
Decision gate	2 weeks	Pilot results reviewed with the Finance Secretary; proceed only if verified findings justify it.	Go / no-go
2. Rollout	6 months	Add Transport, Excise, JBVNL and land records; all 24 districts; officer training; security audit; leadership dashboard.	Statewide platform
3. Run	Ongoing	Rule and rate updates as notifications change; monthly findings and recovery report.	Monthly recovery report

Optional: a municipal edition (holding, water and trade tax) already exists from our earlier Jharkhand work and can be added for Ranchi and two other urban local bodies.

6 BENEFITS

For the Government

- Revenue the state is already owed, found while it can still be recovered rather than years later in audit
- One view of every taxpayer across departments
- Officers' time directed to the highest-value, best-evidenced cases
- Audit readiness: every figure traceable to its record and rate
- Evidence for policy: what a rate or threshold change would yield

For honest taxpayers and businesses

- Fewer blanket inspections; scrutiny targeted at real discrepancies
- A level playing field against under-reporting competitors
- Clear, record-based reasons whenever a case is raised

SCALE, NOT A FORECAST

If verified recoveries reached just 0.1% of own tax revenue (about ₹46 crore a year), that would be about fifty times the platform's full one-time cost of ₹68–92 lakh. How much leakage exists in Jharkhand today is not known by anyone; measuring it on real records is the purpose of the pilot.

7 INDICATIVE COST

Component	Low	High
Pilot (Phases 0–1): adaptation to Jharkhand, 3 departments, 3 districts	₹28 lakh	₹38 lakh
Statewide rollout (Phase 2): 4 more departments, 24 districts, training, security audit	₹40 lakh	₹54 lakh
One-time total	₹68 lakh	₹92 lakh
Infrastructure per year (hosting, storage, AI usage)	₹4 lakh	₹8 lakh
Operations & maintenance per year (support, rule and rate updates, helpdesk)	₹12 lakh	₹18 lakh
Recurring total per year	₹16 lakh	₹26 lakh
Optional: municipal edition for 3 urban local bodies	₹15 lakh	₹22 lakh

Cost assumptions

- The platform already exists; costs cover adaptation to Jharkhand's statutes and registers, integration, deployment, training and support, not development from scratch.
- Delivery cost of ₹1.1–1.4 lakh per person-month for a Kolkata-based team with on-site presence in Ranchi.
- The rollout is paid only if the Government proceeds after the pilot's decision gate.
- If hosted in the Jharkhand State Data Centre (JAP-IT), most infrastructure cost is absorbed by the state.
- All amounts exclude GST and are indicative (±20%) until scoped jointly with the Finance Department.

8 KEY RISKS AND SAFEGUARDS

Risk	How we handle it
Departments reluctant to share data	Read-only access under a Finance Department order; data stays in each department; departments see their own findings first.
Limited access to GST data held by GSTN	Start with the returns and e-way bill data available to the Commercial Taxes Department; national analytics outputs are used alongside, not replaced.
Records do not share a common key	PAN and GSTIN where available; name, address and mobile matching elsewhere, with every inferred link graded and shown as a lead.
Too many false alarms	Evidence separated from leads; ranked lists; officers verify before any action; thresholds tuned on pilot results.
Rates or rules change	One signed-off rate table; updates coded within two weeks of notification.
Privacy and misuse	Role-based access, audit logs, State Data Centre hosting, DPDP Act 2023 compliance.

9 HOW WE WILL WORK WITH THE GOVERNMENT

Start small, prove value	A 4-month pilot on real records in three districts; statewide rollout only after a decision gate.
Transparent procurement	Engagement through GeM, JAP-IT or an empanelled system integrator.
Data stays with the state	Read-only connections; hosting in the Jharkhand State Data Centre or a government-approved Indian cloud.
No vendor lock-in	Open-source technology, source code in escrow, full documentation and training for departmental officers.
Hindi and English	All screens, reports and the assistant in Hindi and English.
Pay for outcomes	Milestones tied to measurable results: verified findings, cases assigned, recoveries tracked.

Sources (public information, accessed September 2026)

1. PRS Legislative Research: Jharkhand Budget Analysis 2026–27 — <https://prsindia.org/budgets/states/jharkhand-budget-analysis-2026-27>
2. Drishti IAS: Jharkhand Budget 2026–27 — <https://www.drishtias.com/state-pcs-current-affairs/jharkhand-budget-202627>
3. CAG: Report No. 4 of 2016, Revenue Sector, Government of Jharkhand — https://cag.gov.in/webroot/uploads/download_audit_report/2016/Jharkhand_Report_No_4_of_2016_Revenue_Sector.pdf
4. CAG: Revenue Sector report on Jharkhand 2017–18 (Report 1 of 2020) — https://cag.gov.in/webroot/uploads/download_audit_report/2017/Report%201%20of%202020%20RS%202017-18,%20Jharkhand,%20Eng-05f6c86e2410dd7.40400418.pdf
5. CAG 2025 report on minor minerals in Jharkhand (news coverage) — <https://thejharkhandstory.co.in/cag-flags-sharp-fall-in-jharkhands-minor-mineral-revenue-points-to-major-lapses/>
6. Business Jharkhand: CAG finds ₹70.92 crore loss, illegal leases (Dec 2025) — <https://www.businessjharkhand.com/2025/12/12/cag-report-revealed-system-failure-and-poor-monitoring-in-jharkhand-the-audit-found-illegal-leases-faulty-auctions-and-70-92-crore-revenue-loss/>
7. The Week: Jharkhand cabinet approves new excise policy (May 2025) — <https://www.theweek.in/wire-updates/national/2025/05/15/ces17-jh-cabinet-excise.html>
8. JIMMS: Jharkhand Integrated Mines & Minerals Management System — <https://mineralsportal.jharkhand.gov.in/aboutJIMMS.aspx>
9. NGDRS / Jharnibandhan: property registration — <https://jharnibandhan.gov.in/>
10. SUDA Jharkhand municipal services portal — <https://municipalservices.jharkhand.gov.in/>
11. Urbanomics: state access to GSTN BIFA and GST Prime — <https://urbanomics.substack.com/p/some-thoughts-on-gst-administration>
12. Karnataka GST analytics portal with IIT Hyderabad — <https://www.casansaar.com/news-GST/karnataka-moves-ahead-with-gst-analytics-portal-development/14457.html>
13. Department of Commercial Taxes, Jharkhand — [https://en.wikipedia.org/wiki/Department_of_Commercial_Taxes_\(Jharkhand\)](https://en.wikipedia.org/wiki/Department_of_Commercial_Taxes_(Jharkhand))

Status descriptions are based on public information up to September 2026 and should be confirmed with the Departments before formal submission.